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Comparing Beta-Blocking Effects of Bisoprolol, Carvedilol and Nebivolol

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Key Words
Bisoprolol · Carvedilol · Nebivolol · Beta-blockers · Pharmacodynamics

Abstract
Objective: Bisoprolol, carvedilol and nebivolol have been shown to be effective in the treatment of heart failure. However, the beta-blocking effects of these drugs have never been compared directly. ***Methods:*** Therefore, we performed a randomized, double-blind, placebo-controlled, cross-over trial in 16 healthy males. Subjects received 10 mg bisoprolol, 50 mg carvedilol, 10 mg nebivolol and placebo on the first morning followed by 5 mg bisoprolol once daily, 25 mg carvedilol twice daily, 5 mg nebivolol once daily and placebo for 1 week. Heart rate and blood pressure were measured at rest and exercise 3 and 24 h following intake of the first dose, and immediately before and 3 hours following intake of the last dose of each drug. In addition, effects of the drugs on nocturnal melatonin release were determined, and quality of life (QOL) was evaluated. ***Results:*** Heart rate at exercise was decreased at 3 h following intake of the first single dose of each drug by bisoprolol (–24%), carvedilol (–17%) and nebivolol (–15%), and at 24 h following intake of the respective last dose of each drug following 1 week of chronic administration by bisoprolol (–14%), carvedilol (12 h; –15%) and nebivolol (–13%) ($p < 0.05$ in all cases). Thus, trough-to-peak-ratios at long-term were as follows: Bisoprolol, 58%; carvedilol (12 h), 85%; nebivolol, 91%. Nocturnal melatonin release was decreased by bisoprolol (–44%, $p < 0.05$) whereas nebivolol and carvedilol had no effect. QOL with carvedilol was slightly but significantly lower than with the other drugs, whereas bisoprolol and nebivolol did not alter QOL. ***Conclusions:*** These data show that peak beta-blocking effects of bisoprolol appear stronger than those of nebivolol and carvedilol. On the other hand, nebivolol exerts the highest trough-to-peak-ratio. However, beta-blocking effects of all the three drugs are similar at trough. Only bisoprolol but neither nebivolol nor carvedilol decreased nocturnal melatonin release, a feature which might cause sleep disturbances. Finally, only carvedilol slightly decreased QOL, whereas nebivolol and bisoprolol did not affect QOL. We conclude that different beta-blockers may exert clinically relevant different effects.

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Introduction

The discussion on potentially different effects of different beta-blockers in the treatment of heart failure is still ongoing, thus being emphasized by the fact that five papers studying four different beta-blockers were published in a single recent issue of the *European Journal of Heart Failure* [1–5].

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In the present study, we investigated three beta-blockers with a Class of Recommendation I and a Level of Evidence A for the treatment of heart failure [6], namely *bisoprolol*, a second generation $beta_1$-selective beta-blocker without additional effects [7, 8]; *carvedilol*, a third generation non-selective beta-blocker with additional vasodilating effects resulting from alpha-blockade [9–11], and *nebivolol*, a third generation $beta_1$-selective beta-blocker with additional NO-mediated vasodilatory effects [12, 13]. All three compounds have been shown to be effective in the treatment of both arterial hypertension and heart failure [7–16]. However, short- and long-term effects of bisoprolol, carvedilol and nebivolol have never been compared directly.

Previously, several differences between different beta-blockers have been described, e.g., it was shown (only in a single study in healthy subjects) that bisoprolol reduced heart rate both at rest and during exercise in healthy subjects, whereas carvedilol was effective only during exercise and failed to produce a significant decrease of heart rate at rest [17], whereas a lack of decrease of heart rate at rest was not shown in any of the previous studies of carvedilol in patients with hypertension, myocardial infarction or heart failure. In addition, it was shown that propranolol and atenolol cause an extensive decrease of nocturnal melatonin release [18, 19], a feature that might cause sleep disturbances, whereas carvedilol lacks this effect [19, 20]. However, the influence of nebivolol and bisoprolol on nocturnal melatonin release never was investigated.

In order to address these unsolved issues, we performed a randomized, double-blind, placebo-controlled, cross-over study in the field of basic clinical cardiovascular pharmacology in 16 healthy males in order to compare beta-blocking efficacy of bisoprolol, carvedilol and nebivolol both at rest and during exercise. In addition, we also determined potential effects of beta-blockade on nocturnal melatonin release and quality of life (QOL).

Methods

The study was performed in 16 healthy males (age 28 ± 6 years, height 183 ± 6 cm, body weight 76 ± 10 kg) according to a randomized, double-blind, placebo-controlled, cross-over protocol. Subjects with any overt disease whatsoever were excluded. All investigations were performed in the morning following an overnight fast. In addition, smoking was prohibited at least 12 h before the onset of all investigations. On the first day, subjects received single oral doses of 10 mg nebivolol, 10 mg bisoprolol, 50 mg carvedilol and placebo, followed by 5 mg bisoprolol once daily, 25 mg carvedilol twice daily, 5 mg nebivolol once daily and placebo, for 1 week. The first single doses of bisoprolol, carvedilol and nebivolol given in our study were equal to those recommended in the well-known large scale clinical trials of these drugs in patients with heart failure (CIBIS II [14], US Carvedilol [15], SENIORS [16]), whereas all the following doses represented 50% of these amounts since our study was performed in healthy volunteers.

In order to guarantee the blinding of all participants, all subjects received two capsules daily: When carvedilol was given, the respective second capsules to be swallowed every day contained 25 mg carvedilol; however, when nebivolol, bisoprolol or placebo was given, the respective second capsules every day contained placebo.

The order of taking the drugs was determined in four parallel groups by random order and achieved by drawing which revealed the following sequences: Subjects 1, 5, 9, 13 – bisoprolol, nebivolol, carvedilol, placebo; subjects 2, 6, 10, 14 – carvedilol, bisoprolol, placebo, nebivolol; subjects 3, 7, 11, 15 – placebo, carvedilol, nebivolol, bisoprolol; subjects 4, 8, 12, 16 – nebivolol, placebo, bisoprolol, carvedilol.

Investigations were performed immediately before and 3 h following intake of the first dose (day 1), 24 h (carvedilol, 12 h) following intake of the first dose (day 2), and 24 h (carvedilol, 12 h) following intake of the last dose (day 8) in order to investigate acute and long-term effects of the drugs as well as trough-to-peak ratios. In this context, 7 days of continuous intake of the drugs appears long enough in order to reach clear steady state in this study in healthy subjects and, herewith, draw basic pharmacologic conclusions on long-term effects.

Hemodynamic Measures

Heart rate (by continuous ECG monitoring) and blood pressure (by sphygmomanometry) were measured at rest (immediately before exercise) and during the last minute of 10 min of exercise on a bicycle ergometer with 70% of mean individual work load (determined by age, height and body weight, mean 156 ± 13 W). These investigations were performed before the intake of the first dose of either drug (baseline) and repeated 3 h following intake of the first dose (this interval was chosen since peak plasma concentrations and peak effects may be expected at this time [7–12]), 24 h (carvedilol, 12 h) following intake of the first dose (day 2), and 24 h (carvedilol, 12 h) following intake of the last dose (day 8). Herewith, we determined baseline values, peak and trough effects following the first single doses of either drug, as well as trough effects during long-term administration of the drugs in order to compare beta-blocking effects of the drugs both at rest and during exercise.

Nocturnal Melatonin Release

During the first and seventh night with each drug, subjects had to sleep in a dark room between 10 p.m. and 6 a.m., and the complete nocturnal urine produced during this time was collected. Urinary aMT6s concentrations were determined using a Bühlmann 6-SMT enzyme-linked immunosorbent assay (Bühlmann Laboratories AG, Allschwil, Switzerland). This validated [20] and commercially available aMT6s ELISA is a competitive immunoassay using an antibody-capture technique with a lower detection limit of 0.8 ng/ml for aMT6s. It is specific for aMT6s and shows no relevant cross-reactivity for melatonin, its metabolites or related compounds (<0.01%). Intra- and inter-assay precision was calculated to be 6.4 and 10.5%, respectively. The known concentration of aMT6s was multiplied by the measured volume of urine to give the total amount of melatonin excreted. Because of the exact timing of urine collection and known urine volume the dilution of urine was negligible.

Table 1. Hemodynamic effects

	Placebo	Bisoprolol	Carvedilol	Nebivolol
HR rest baseline	66 ± 5	65 ± 3	66 ± 7	66 ± 6
Syst BP rest baseline	116 ± 8	118 ± 10	118 ± 9	118 ± 10
Diast BP rest baseline	67 ± 4	67 ± 3	68 ± 5	68 ± 5
HR exercise baseline	161 ± 12	158 ± 13	158 ± 15	160 ± 14
Syst BP exercise baseline	178 ± 15	176 ± 17	178 ± 16	174 ± 15
Diast BP exercise baseline	69 ± 7	67 ± 7	66 ± 6	66 ± 8
HR rest 3 h	63 ± 7	52 ± 8* (–21%)	57 ± 5* (–12%)	56 ± 6* (–15%)
Syst BP rest 3 h	115 ± 9	106 ± 9* (–9%)	105 ± 10* (–11%)	109 ± 8* (–8%)
Diast BP rest 3 h	68 ± 3	61 ± 5	63 ± 5	65 ± 5
HR exercise 3 h	162 ± 13	123 ± 10* (–24%)	131 ± 14* (–17%)	135 ± 14* (–15%)
Syst BP exercise 3 h	178 ± 15	158 ± 17* (–11%)	160 ± 17* (–9%)	162 ± 19* (–9%)
Diast BP exercise 3 h	68 ± 8	65 ± 8	65 ± 6	67 ± 7
HR rest 24 h	61 ± 7	51 ± 6* (–22%)	51 ± 5* (–23%)	56 ± 7* (–15%)
Syst BP rest 24 h	115 ± 10	108 ± 8* (–8%)	106 ± 8* (–10%)	110 ± 11* (–7%)
Diast BP rest 24 h	66 ± 5	62 ± 4	60 ± 5	62 ± 6
HR exercise 24 h	160 ± 13	132 ± 11* (–16%)	134 ± 14* (–15%)	138 ± 12* (–14%)
Syst BP exercise 24 h	176 ± 15	168 ± 19* (–5%)	163 ± 16* (–8%)	169 ± 20* (–3%)
Diast BP exercise 24 h	67 ± 8	66 ± 7	64 ± 6	65 ± 7
HR rest 7 days	62 ± 6	55 ± 5* (–15%)	56 ± 5* (–15%)	57 ± 7* (–14%)
Syst BP rest 7 days	115 ± 7	114 ± 7* (–3%)	112 ± 6* (–5%)	111 ± 7* (–6%)
Diast BP rest 7 days	65 ± 4	61 ± 5	63 ± 5	62 ± 6
HR exercise 7 days	157 ± 14	139 ± 16* (–12%)	135 ± 13* (–15%)	137 ± 11* (–15%)
Syst BP exercise 7 days	176 ± 17	170 ± 19* (–3%)	164 ± 16* (–8%)	166 ± 17* (–5%)
Diast BP exercise 7 days	66 ± 8	63 ± 7	65 ± 6	63 ± 7

Heart rate (HR) and systolic blood pressure (syst BP) at baseline, at 3 h and at 24 h (carvedilol 12 h) following oral intake of the first (single) doses of bisoprolol (10 mg), carvedilol (50 mg) and nebivolol (10 mg), and on the eighth day at 24 h (carvedilol 12 h) following the last doses of 7 days of chronic administration of bisoprolol (5 mg once daily), carvedilol (25 mg twice daily) and nebivolol (5 mg once daily).

Effects on HR and systolic BP vs. baseline are given in parentheses (%). Means ± SD; n = 16; * $p < 0.05$ (compared to baseline).

Quality of Life

Subjects had to fill out a questionnaire on QOL (developped especially for the present study) following the first and seventh day of intake of either drug in the fields of general well-being, performance, vertigo, erectility, libido and quality of sleep, by grading QOL between 0 (= worst) and 10 (= best). Finally, all these QOL measures were taken together for each drug and compared as 'total QOL'.

Statistical Methods

Results are given as means ± SD unless otherwise indicated. Significances of differences within groups were calculated by Repeated Measures ANOVA (Repeated Measures ANOVA on Ranks whenever data were not normally distributed) throughout the study, post-hoc analyses were performed using the Student-Newman-Keuls test, and $p < 0.05$ was considered statistically significant.

Ethical Considerations

All investigations conformed with the principles outlined in the Declaration of Helsinki, the study was approved by the Ethical Committee of the Medical University of Graz, Austria (protocol number 11-039 ex 00/01), and all participants gave written informed consent.

Results

Hemodynamic Effects

Hemodynamic effects are shown in table 1 and figure 1. Compared to baseline, heart rate during exercise was decreased at 3 h following intake of the first dose of either drug by bisoprolol (–24%), carvedilol (–17%) and

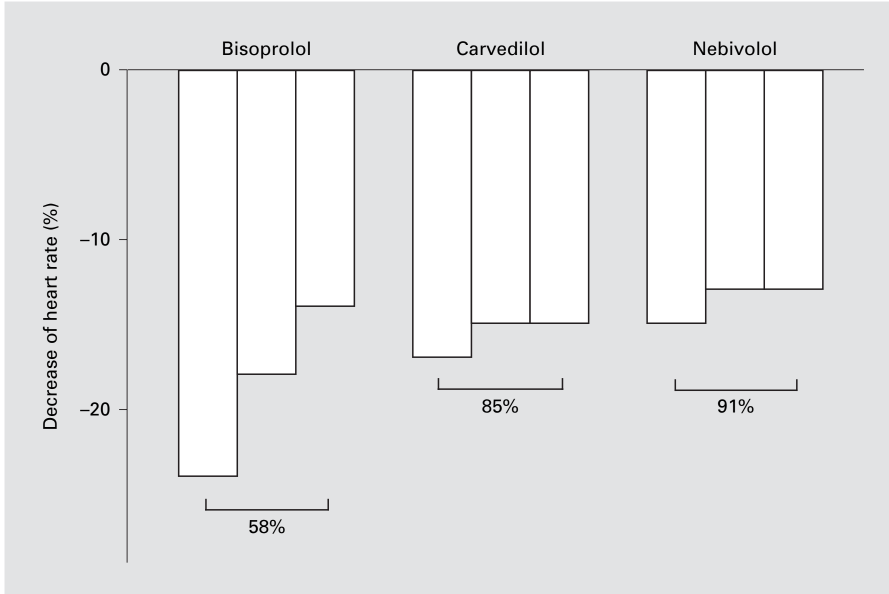

Fig. 1. Effects of bisoprolol, carvedilol and nebivolol on heart rate during exercise and trough-to-peak-ratios: First column of each drug = 3 h following the first (single) dose. Second column of each drug = 24 h*) following the first (single) dose. Third column of each drug = 24 h*) following the last dose of 7 days of chronic treatment. Trough-to-peak ratios are given as means; n = 16; * Carvedilol = 12 h.

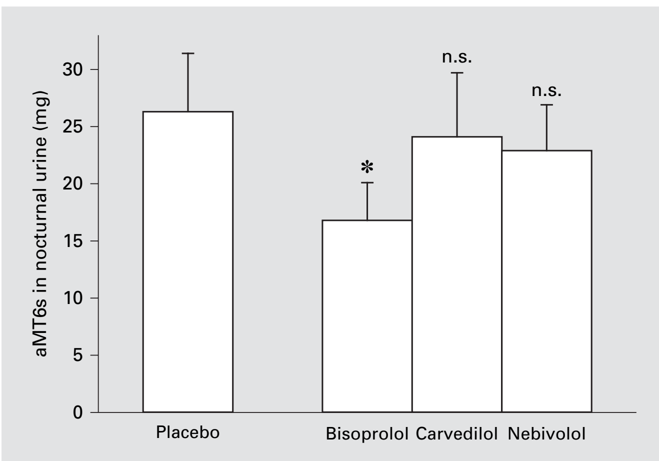

Fig. 2. Melatonin release (expressed as total aMT6s in nocturnal urine) during the first night following oral intake of 10 mg bisoprolol, 50 + 25 mg carvedilol, 10 mg nebivolol and placebo. Nocturnal melatonin release was decreased only by bisoprolol (–44%, $p < 0.05$) whereas nebivolol and carvedilol had no effect. Means ± SEM; n = 16; * $p < 0.05$; n.s. = not significant (compared to placebo).

nebivolol (–15%) (the effect of bisoprolol being higher than that of nebivolol, $p < 0.05$); at 24 h following intake of the first single doses by bisoprolol (–18%), carvedilol (12 h!; –15%) and nebivolol (–13%) (no significant differences between the drugs); 24 h following intake of the respective last dose at the end of 1 week of chronic administration by bisoprolol (–14%), carvedilol (12 h!; –15%) and nebivolol (–13%) (no significant differences between the drugs). However, the effects of all the drugs on heart rate compared to placebo were significant ($p < 0.05$ in all cases).

Trough-to-peak-ratios (fig. 1) of the effects of the drugs on heart rate at exercise at 24 h following oral intake of the respective first single doses (carvedilol 12 h following intake of a second dose of 25 mg) were as follows: Bisoprolol, 76%; carvedilol (12 h!), 89%; nebivolol, 87%; at 24 h following intake of the respective last doses following 7 days of chronic administration: Bisoprolol, 58%; carvedilol (12 h!), 85%; nebivolol, 91%. In both cases, trough-to-peak-ratios of carvedilol (12 h!) and nebivolol (24 h!) were similar but significantly higher than those of bisoprolol (24 h!) ($p < 0.05$ in all cases).

All the drugs significantly decreased systolic blood pressure both at rest and exercise at 3 and 24 h following intake of the respective first single drugs as well as 24 h following intake of the respective last single drugs following 7 days of chronic administration ($p < 0.05$ in all cases). None of the drugs had any significant effect on diastolic blood pressure neither at rest nor at exercise in the present study with healthy subjects.

Nocturnal Melatonin Release

Effects of the drugs on nocturnal melatonin release are shown in figure 2. Total amounts of aMT6s in nocturnal urine (1st and 7th night) were as follows: Placebo, 26.9 ±

Table 2. Quality of life (QOL)

QOL	Placebo	Nebivolol	Bisoprolol	Carvedilol	p value
Day 1					
General well-being	8.6 ± 1.4	8.9 ± 1.1	8.2 ± 1.6	7.6 ± 1.6	0.047
Performance	8.1 ± 1.5	8.3 ± 1.7	7.7 ± 1.7	7.1 ± 1.9	0.153
Vertigo	9.4 ± 1.3	9.6 ± 0.8	9.2 ± 1.2	9.6 ± 0.8	0.673
Erectility	8.5 ± 1.6	8.6 ± 1.6	8.2 ± 1.4	7.9 ± 1.8	0.307
Libido	8.4 ± 1.9	8.6 ± 1.9	8.2 ± 1.5	8.1 ± 1.7	0.515
Quality of sleep	8.8 ± 1.7	8.6 ± 2.0	8.3 ± 2.3	8.3 ± 2.0	0.603
Day 7					
General well-being	8.5 ± 1.4	8.2 ± 2.4	8.5 ± 2.1	7.4 ± 2.1	0.045
Performance	8.3 ± 1.5	8.3 ± 1.7	7.9 ± 1.9	6.9 ± 2.3	0.052
Vertigo	9.6 ± 1.3	9.4 ± 1.0	9.6 ± 0.7	9.2 ± 1.4	0.506
Erectility	8.4 ± 1.7	7.9 ± 1.9	8.3 ± 1.9	7.8 ± 2.3	0.588
Libido	8.5 ± 1.6	7.9 ± 2.2	8.2 ± 2.1	7.9 ± 2.4	0.592
Quality of sleep	8.8 ± 1.4	7.8 ± 2.0	8.7 ± 1.4	8.3 ± 2.3	0.049
Total QOL	8.6 ± 0.4	8.5 ± 0.6	8.4 ± 0.5	8.0 ± 0.9	<0.001

Quality of life (QOL) on the first and seventh day of oral intake of nebivolol, bisoprolol, carvedilol, and placebo, with QOL being graded between 0 (= worst) and 10 (= best). All these QOL measures taken together for each drug are given as total quality of life. In this category, QOL with placebo, nebivolol and bisoprolol being superior to that with carvedilol ($p < 0.05$ in all cases).

21.4 and 26.3 ± 22.4 mg; bisoprolol, 14.9 ± 20.0 and 16.8 ± 13.4 mg (–44% and –36% compared to placebo, $p < 0.05$ in both cases); carvedilol, 22.5 ± 20.0 and 24.1 ± 22.3 mg (–16 and –8% compared to placebo, n.s.); nebivolol, 21.9 ± 17.0 and 22.9 ± 16.1 mg (–19 and –13% compared to placebo, n.s.).

Quality of Life

Effects of the drugs on QOL are shown in table 2. Total QOL with carvedilol (8.0 ± 0.8) was slightly but significantly lower than that with placebo (8.6 ± 0.4), nebivolol (8.5 ± 0.6) and bisoprolol (8.4 ± 0.5) ($p < 0.05$ in all cases). General well-being with nebivolol on day 1 (8.9 ± 1.1) was significantly better than with carvedilol (7.6 ± 1.6) ($p < 0.05$). In all the other subgroups of QOL (performance, vertigo, erectility, libido and quality of sleep) there were no significant differences between bisoprolol, carvedilol, nebivolol and placebo.

Discussion

These data indicate that beta-blocking effects of bisoprolol, carvedilol and nebivolol show clinically relevant differences in the fields of hemodynamic effects, effects on nocturnal melatonin release and QOL.

Regarding hemodynamic effects, bisoprolol caused the highest acute decrease of heart rate, but its trough-to-peak-ratios declined with chronic administration. Carvedilol showed a somewhat weaker acute decrease of heart rate, but trough-to-peak-ratios remained constant during long-term treatment, perhaps in part due to the fact that the dosing intervals with carvedilol were only 12 h (according to recommendations for the clinical use of carvedilol, subjects received 25 mg carvedilol twice daily, thus being the same amount of the drug as the first dose whereas the daily doses of bisoprolol and nebivolol were only half that of the respective first doses of these drugs), whereas the dosing intervals of bisoprolol and nebivolol were 24 h. Nebivolol showed the lowest acute decrease of heart rate, but trough-to-peak-ratios did not decrease with ongoing continuous administration, with nebivolol, finally showing the highest trough-to-peak-ratios. However, beta-blocking effects of all the three drugs were similar at trough, both at 24 h (carvedilol, 12 h) following intake of the first dose as well as 24 h (carvedilol, 12 h) following intake of the last dose of each drug following one week of chronic administration. Therefore, all the investigated drugs showed similar amounts of decrease of heart rate during exercise even at 24 h (carvedilol, 12 h) following drug intake, bisoprolol showing the highest acute decrease of heart rate, and nebivolol having the highest trough-to-peak-ratio during long-term administration. However, it has to be emphasized that the first doses used in the present study were the double of those administered long-term (except with carvedilol), thus being a limitation of the present study. Therefore, trough-to-peak-ratios found in the present study should be interpreted with care.

Systolic blood pressure was decreased significantly and to a similar extent by all our drugs both at rest and exercise, whereas none of them had any significant effect on diastolic blood pressure in this study in healthy volunteers.

It was shown previously that beta-blockers such as propranolol and atenolol markedly decrease nocturnal melatonin release [18, 19], this finding not being unexpected since melatonin release is triggered by stimulation of beta$_1$-receptors in the pineal gland [22, 23]. However, in the present study nocturnal melatonin release was decreased only by bisoprolol whereas nebivolol and carvedilol had no effect. One reason for this unpredicted lack of effect of carvedilol on nocturnal melatonin release might be that the decrease of blood pressure caused by the alpha-blocking effect of carvedilol might cause a reflex increase in sympathetic drive, thus counteracting at least in part the beta-blocking effect of this drug on the pineal gland. Indeed, it was shown previously that carvedilol did not cause clinically relevant beta-blockade at rest in healthy subjects, with even increasing heart rates at rest comparing increasing doses of carvedilol, thus a feature quite unexpected with a beta-blocker [17, 24]. Since nocturnal melatonin release was measured at sleep, i.e., at a situation of maximum rest, clinically relevant beta-blockade of carvedilol might have been too weak to cause a decrease of nocturnal melatonin release. On the other hand, the reason why nebivolol failed to decrease nocturnal melatonin release cannot be explained from the data of the present study. One can only speculate that the chemical structure of nebivolol, which is markedly different from that of all other beta-blockers actually used in clinical practice, might be a reason for the lack of effect of this drug on nocturnal melatonin release. However, the fact that both carvedilol and nebivolol did not decrease nocturnal melatonin release might be one reason why these third generation beta-blockers rarely cause sleep disturbances as a side effect. On the other hand, this conclusion cannot be drawn from another finding of the present study, namely that quality of sleep with bisoprolol was not worse than with the other drugs, and nebivolol even showed a slight (although non-significant) tendency towards a lower quality of sleep than bisoprolol on day 7. However, none of the three beta-blockers investigated in the present study significantly worsened quality of sleep. Taken together, it has to be emphasized that these explanations are still speculative since, to our knowledge, the clinical relevance of melatonin on quality of sleep in patients receiving beta-blockers has never been investigated in clinical trials performed to address this issue.

Total QOL with nebivolol and bisoprolol did not differ from that obtained with placebo. However, QOL with carvedilol appeared somewhat worse than that with placebo, bisoprolol and nebivolol ($p < 0.05$). In addition, general well-being was best with nebivolol, thus not differing from placebo and bisoprolol. On the other hand, general well-being with carvedilol appeared somewhat worse than with nebivolol ($p < 0.05$). In all the other subgroups of QOL investigated in the present study (performance, vertigo, quality of sleep, even libido and erectility!) there were no differences between bisoprolol, carvedilol, nebivolol and placebo. However, these findings obtained in the present study with healthy males should be interpreted with care.

One potential reason for the lower QOL with carvedilol in the present study might be the alpha-blocking effect of this drug. However, according to the rather low number of participants in the present study, these slight (but statistically significant) differences between the QOL with carvedilol on the one hand and bisoprolol and particularly nebivolol on the other hand might possibly be a play of chance as well and, therefore, should be interpreted with care.

In addition to the different effects of beta-blockers obtained in the present study, it was shown recently that pharmacokinetic properties of third generation beta-blockers also differ significantly from those of first and second generation beta-blockers: It was shown previously that plasma concentrations of propranolol, atenolol and bisoprolol markedly increase during exercise [25–29]. The reason for this increase might be the fact that propranolol and atenolol are taken up into, stored in and released from adrenergic cells during exercise together with epinephrine and norepinephrine, thus causing an increase of plasma concentrations of these drugs together with those of epinephrine and norepinephrine during exercise [30–35]. However, it was shown recently that plasma concentrations of carvedilol [36, 37] and nebivolol [29] do not increase during exercise, thus being a further major difference between third generation beta-blockers and those of the first and second generation [38].

In conclusion, different beta-blockers (particularly those of the third generation) may exert clinically relevant pharmacodynamic differences (as well as differences from the pharmacokinetic point of view). Therefore, beta-blockers should be seen and used as individual substances on their own rather than as 'all equal' members of one class of drugs.

Since the last decade, recommendations in the treatment of arterial hypertension have focused on the patient as a whole, including all additional diseases and risk factors, rather than only on blood pressure alone. As a consequence of the obvious differences between beta-blockers shown in the present study, a similar therapeutic strategy with beta-blockers might make sense, i.e., to focus the choice of a special beta-blocker on additional diseases and risk factors of the individual patient according to the particular features of each drug. However, further clinical studies are needed in order to find out whether or not such kind of policy might further improve clinical outcome and QOL.

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